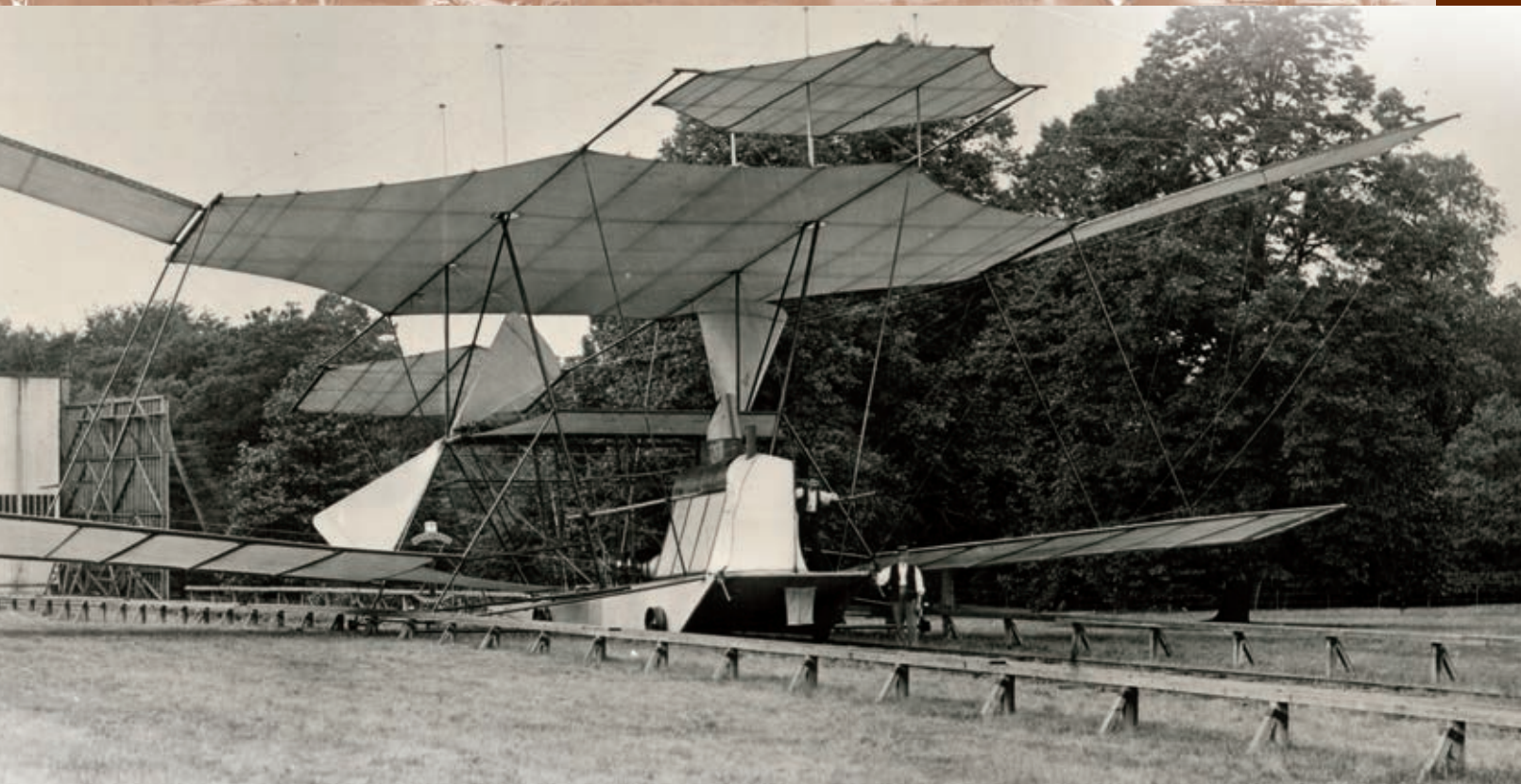




ascertained, was to skim the ground at a height of around 8in (20cm) for a distance of 165ft (50m). This could not be called controlled, sustained flight, but it was a start. To fund further experiments, Ader turned to the French Ministry of War, which was eager to explore any “secret weapon” that might give France an edge over its neighbor, Germany. Armed with the first military budget for airplane development, Ader built a twin-engined aircraft, the *Avion III*. But when tested in front of military observers in October 1897, it failed to get off the ground. Funding was cut off and Ader’s experiments came to an end.

The most prominent advocate of the power-centered approach to flight was Sir Hiram Maxim, the American-born inventor of the Maxim machine gun. He wrote that it was “neither necessary nor practical to imitate the bird... Give us a motor and we will very soon give you a successful flying machine.” In the 1890s, Maxim devoted a large part of his fortune to developing a huge biplane on his estate in Kent, England. Maxim did not intend this giant to fly in any conventional sense. It sat on a test track consisting of two sets of rails. The airplane was to run on

the lower rails while gathering speed for takeoff; the upper rails were to prevent it from rising into free flight, which would inevitably result in a crash. In July 1894, Maxim’s machine rose from its rails after accelerating to 42mph (67kph), only to collide with the upper rails, sustaining heavy damage. His experiments progressed no further.

Maxim, Ader, and other power-centered flight experimenters had given little or no thought to how they would actually fly their machines should

LARGE-SCALE BIPLANE

This test rig, built by the American-born Hiram Maxim in the 1890s, was a truly huge aircraft. With a wingspan of 107ft (33m) and weighing more than 3.9 tons (3.6 tonnes), it took two 180hp steam engines, each driving a propeller 18ft (5.5m) in diameter, to drive it along its restraining rails.

SOCIETY VISIT

Hiram Maxim (reclining, center) poses with members of the British Aeronautical Society alongside his massive steam-driven biplane at Baldwyns Park in Kent, England.

